TCS Logic Validation [1 .. 5] — Pass

Test Case Results

Pass	5		100%
Fail	0	0%	
Total	5		100%

Measurement was performed with simulation running as fast as possible.

General Information

Test Begin: 10/06/2026 09:18:01 AM +05:30
Test End: 10/06/2026 09:18:01 AM +05:30
Duration: 00:00:00

Automated validation of the Traction Control System logic and fail-safes.

Tester

Login Name: admin

Test Setup

Version:	CANoe.CAN.LIN.Ethernet.BasicEthernet 19.5.44 (SP5) /pro
Configuration:	C:\Elc23019\CAN_500kBaud_2ch\CAN_500kBaud_2ch.cfg
Database Elc23019 on CAN1:	C:\Elc23019\CAN_500kBaud_2ch\Elc23019.dbc (2026-09-09 12:37:12)
Test Module Name:	TCS_TestCase
Test Module File:	C:\Elc23019\CAN_500kBaud_2ch\TCS_TestCase.can
Last modification of Test Module File:	2026-10-05, 14:55:51
Computer Name:	202526AUGC0028
Nodelayer Module CANoeILNLVector:	CANoe Interaction Layer NL (Vector) Version 3.30.156 (Aug 5 2025 12:33:58) , c:\ProgramData\Vector\AddOn Packages\VectorBase AddOn\vModules\CANoe_14_0\Exec64\CANoeILNLVector.dll

Table of Contents

	TCS Logic Validation [1 .. 5]	Pass
	Core Intervention Tests [1 .. 4]	Pass
1	**TC_01_NormalDriving_NoIntervention**	Pass
2	**TC_02_IceSlip_TriggersIntervention**	Pass
3	**TC_03_TCSOffSwitch_DisablesIntervention**	Pass
4	**TC_04_IgnitionOff_BlocksIntervention**	Pass
	Diagnostic Tests [5]	Pass
5	**TC_05_WheelSensorFault_SetsFlag**	Pass

TCS Logic Validation [1 .. 5] — Pass

Main Part

Time Stamp	Ident	Title	Verdict
8637.500000		**Core Intervention Tests** [1 .. 4]	Pass
8643.000000		**Diagnostic Tests** [5]	Pass

Core Intervention Tests [1 .. 4] — Pass

Validating slip thresholds and TCS behavior

Test Group Begin: 10/06/2026 09:18:01 AM +05:30
Test Group End: 10/06/2026 09:18:01 AM +05:30
Duration: 00:00:00

Main Part

Time Stamp	Ident	Title	Verdict
8637.500000		1. **TC_01_NormalDriving_NoIntervention**	Pass
8639.000000		2. **TC_02_IceSlip_TriggersIntervention**	Pass
8640.500000		3. **TC_03_TCSOffSwitch_DisablesIntervention**	Pass
8642.000000		4. **TC_04_IgnitionOff_BlocksIntervention**	Pass

1. TC_01_NormalDriving_NoIntervention — Pass

Test Case Begin: 10/06/2026 09:18:01 AM +05:30
Test Case End: 10/06/2026 09:18:01 AM +05:30
Duration: 00:00:00

Main Part

Time Stamp	Test Step	Title	Verdict
8638.000000		Waited for 500 ms.	
8639.000000		Waited for 1000 ms.	
8639.000000		Slip stayed below 15 on Dry at 50% throttle — correct	Pass

2. TC_02_IceSlip_TriggersIntervention — Pass

Test Case Begin: 10/06/2026 09:18:01 AM +05:30
Test Case End: 10/06/2026 09:18:01 AM +05:30
Duration: 00:00:00

Main Part

Time Stamp	Test Step	Title	Verdict
8639.500000		Waited for 500 ms.	
8640.500000		Waited for 1000 ms.	
8640.500000		Slip exceeded 15 on Ice at 100% throttle — correct	Pass
8640.500000		TCS ACTIVE LED correctly turned on	Pass

3. TC_03_TCSOffSwitch_DisablesIntervention — Pass

Test Case Begin: 10/06/2026 09:18:01 AM +05:30
Test Case End: 10/06/2026 09:18:01 AM +05:30
Duration: 00:00:00

Main Part

Time Stamp	Test Step	Title	Verdict
8641.500000		Waited for 1000 ms.	
8642.000000		Waited for 500 ms.	
8642.000000		TCS ACTIVE LED correctly turned off after TCS OFF switch	Pass

4. TC_04_IgnitionOff_BlocksIntervention — Pass

Test Case Begin: 10/06/2026 09:18:01 AM +05:30
Test Case End: 10/06/2026 09:18:01 AM +05:30
Duration: 00:00:00

Main Part

Time Stamp	Test Step	Title	Verdict
8643.000000		Waited for 1000 ms.	
8643.000000		TCS correctly stayed inactive with Ignition Off	Pass

Diagnostic Tests [5] — Pass

Validating fault flags

Test Group Begin: 10/06/2026 09:18:01 AM +05:30
Test Group End: 10/06/2026 09:18:01 AM +05:30
Duration: 00:00:00

Main Part

Time Stamp	Ident	Title	Verdict
8643.000000		5. **TC_05_WheelSensorFault_SetsFlag**	Pass

5. TC_05_WheelSensorFault_SetsFlag — Pass

Test Case Begin: 10/06/2026 09:18:01 AM +05:30
Test Case End: 10/06/2026 09:18:01 AM +05:30
Duration: 00:00:00

Main Part

Time Stamp	Test Step	Title	Verdict
8643.500000		Waited for 500 ms.	
8643.500000		Wheel sensor fault flag correctly set	Pass